

Towards an Expanded Eyebbox for a Wide-Field-of-View Augmented Reality Near-eye Pinlight Display with 3D Pupil Localization

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Figure 1: (a) 3D-printed wearable prototype. (b) We use a 6 mm lens to focus at 0.3 m, 1.3 m, and 5.0 m respectively. Real-world objects and augmentation imagery are visible at each focused depth. (c) We use a 4 mm lens and control its rotation with a motorized rotation stage. The camera rotates through 24°, and all imagery remains visible.

ABSTRACT

A head-worn optical-see-through near-eye display (NED) is crucial for augmented reality (AR), enabling simultaneous perception of virtual and real imagery. Current AR NEDs face trade-offs among field of view (FOV), eyebox size, and form factor. We designed an enhanced pinlight AR NED using a novel approach to capture the pupil's 3D location and compute a modulation pattern on the display. In our design, an eye-tracking camera rig captures stereoscopic views to calculate a display pattern, modulating light beams for precise imagery. Our compact prototype achieves a wide FOV of 79° in horizontal, a large eyebox, and experimental results confirm its spatial and temporal consistency.

Index Terms: Augmented Reality, Near-eye Display, Pinlight Display, Eye Tracking.

1 INTRODUCTION

It is frequently asserted in academia and industry that augmented reality (AR) will be a game changer for next-generation computing platforms. Optical-see-through near-eye displays (NEDs) for AR are capable of combining real and virtual imagery such that the user can simultaneously perceive both [4]. As of today, several optical see-through AR NED products are available on the market,

including Microsoft HoloLens 2, Magic Leap 2, and others. Despite intense effort in academia and industry to advance optical-see-through AR NEDs, all existing technologies and products involve trade-offs between a few key metrics: field of view (FOV), eyebox size, form factor, and resolution [1]. Such compromises constrain existing AR NEDs from being widely accepted, especially for all-day usage by the general public [5].

Pinlight displays show promise for optical-see-through AR NEDs requiring a wide FOV and a compact form factor [2]. The pinlight display design [3] solves the light-field mismatching problem through the use of pinholes. However, pinlight displays offer a limited eyebox. To address this shortcoming, we have developed a novel method to expand the eyebox through eye tracking. In our design, the continually captured pupil position is used to control in real-time the pattern displayed by the spatial light modulator, such that the eye is able to perceive uniform, high-quality images.

2 OUR METHODS

2.1 Design

Our proposed design contains three main components: a pinlight plane, a transparent modulation plane, and an eye-tracking camera rig. The pinlight plane is a transparent light-guided panel with a dense array of light guide dots. The LED light source is mounted on the edge of the pinlight plane to generate a series of point light sources. The pinlight plane emits light that traverses the modulation plane (e.g., LCD), with each modulation element on the modulation plane corresponding to a beam of light emitted from the point light source on the pinlight plane. The modulation plane modulates the intensities of the light beams that traverse it before the light beams enter the user's eye, creating the perceived image on the retina. The eye-tracking camera rig is located at the bottom of the pinlight AR NED and oriented towards the user's eye. It is used to determine the pose of the user's eyeball in a 3D space, i.e. position, viewing direction, as well as pupil size. The eyeball pose is used to calculate the pattern displayed on the modulation plane.

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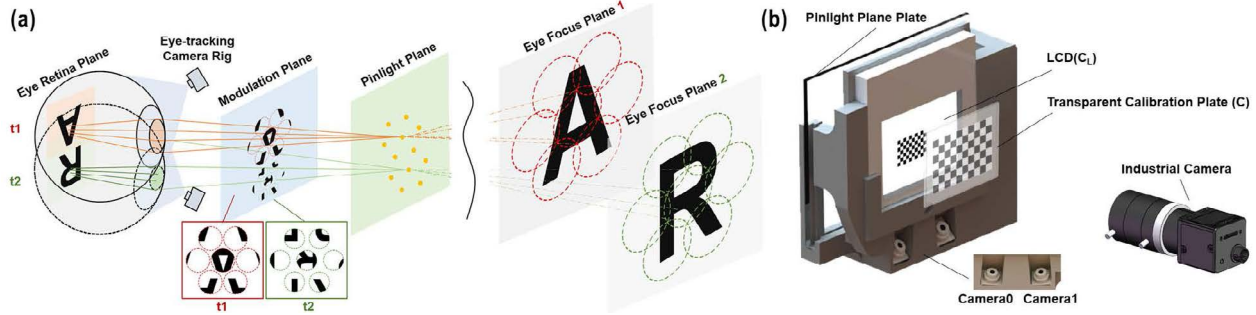


Figure 2: (a) The principle of our proposed pinlight AR NED with pupil localization. (b) Schematic setup for calibration between the modulation plane, pinlight plane, and eye-tracking camera rig.

2.2 Modulation Calculation Using Eye-Tracking

Figure 2 (a) shows how the modulation pattern changes when the eyeball pose changes. At time t_1 , the human eye looks forward, focusing at a close depth. After the eye-tracking camera rig captures the relevant information including the 3D pupil location and size, the corresponding modulation pattern is calculated to allow the eye to see image "A" on the eye focus plane 1. At time t_2 , the eyeball rotates an angle with the pupil size, position, and accommodation depth also changed. The modulation plane then loads the corresponding modulation pattern, enabling the eye to see image "R" on the eye focus plane 2. Thus, whenever the state of the eyeball changes, a suitable modulation pattern is calculated in response and is updated on the modulation plane to modulate the light beams, thereby effectively expanding the eyebox.

2.3 Prototype Implementation

The pinlight plane is fabricated from a 2 mm thin acrylic sheet with laser-drilled tiny divots, and is illuminated by LEDs on its top edge. The light from the LEDs is transmitted throughout the acrylic sheet and scattered at the divots, forming small bright spots. The spatial light modulator consists of a 5 mm thin transparent LCD panel located 14 mm in front of the pinlight plane. For a typical eye-to-LCD distance of 20 mm, the resulting horizontal FOV is 79° while the vertical FOV is 63° . The eye-tracking camera rig contains two cameras per eye, which capture stereoscopic images of the eyes.

To calibrate the transformations between the eye-tracking camera rig, the pinlight plane and the LCD, we employ an industrial camera and a transparent checkerboard C (Fig. 2 (b)). The following describes the calibration procedure. 1) We compute the pose of C in the Camera0 coordinate system O_E by stereo camera calibration. 2) We compute the pose of C in the industrial camera's coordinate system O_I . This also provides the transformation between O_E and O_I . 3) Using the checkerboard C_L displayed on the LCD, we can obtain the transformation between O_I and the LCD coordinate system O_L , and thus also the transformation between O_E and O_L . 4) The industrial camera detects the dots on the pinlight plane, whereupon we compute the transformation between O_L and the pinlight plane coordinate system O_P . Using these calibration results at runtime, we can calculate the modulation pattern from the eyeball pose information in O_L coordinates as described.

3 EXPERIMENTS

We use an industrial camera and a motorized rotation stage to simulate the eyeball. The industrial camera can be fitted with either a 4 mm or a 6 mm focal length lens. Lens focus can be adjusted to simulate the eye accommodating at different depths. We use an Arduino kit to control the motorized rotation stage that simulates eye motion. In Fig. 1(a), the industrial camera is mounted with a 6 mm lens which has a shallower depth of field and comes with a 3 mm aperture, and it focuses at different depths. Sharp imageries at

0.3 m, 1.3 m, and 5 m can be observed respectively. In Fig. 1(b), we switch to the wider, deeper-focus 4 mm lens with a 2 mm aperture and control its rotation. The rotation range is from -12° to 12° , and a clear synthetic image can be seen throughout that entire range as the system responds to the modified "eye" pose. It can be seen that the virtual image remains in its original position in the real environment despite the rotation of the camera.

4 CONCLUSION AND FUTURE WORK

We have developed a novel design to enhance pinlight AR NEDs by expanding the eyebox via 3D pupil localization. A novel calibration method was deployed to calibrate the eye-tracking stereo camera rig, the pinlight plate, and the modulation plane. An eyeglass-style prototype was implemented with our design. We conducted experiments to verify the effectiveness of our proposed design to expand the eyebox for pinlight AR NEDs.

Our current design supports only monochrome imagery. A rainbow effect and tiny barriers can be observed on the perceived image which are caused by the LCD panel used as the modulation plane. Therefore, we plan to improve our design to support multiple colors and reduce the rainbow and barrier image artifacts. We also plan to improve our calibration methods towards efficient 3D pupil localization and perspective projection. We will also conduct comprehensive user studies to evaluate our enhanced design.

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